

First Part of our outline

① Vector and geometry of space

① Rectangular coordinate system:-

A rectangular coordinate system, also known as a cartesian coordinate system, is a way to represent points in a plane or space using order pairs.

In 2-D

x, y

In 3-D

x, y, z

Distance formula between two points

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

② Cylindrical coordinate system:-

A cylindrical coordinate system is a way to represent point in 3D space using three coordinates r, θ and z .

Conversion to cartesian coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$\text{Conversion from cartesian coordinates} \\ r = \sqrt{x^2 + y^2} \quad \left| \quad \theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \right| \quad z = z$$

Example:- Convert the point $(3, 4, 5)$ in cartesian coordinates to cylindrical coordinates.

$$r = \sqrt{x^2 + y^2} = \sqrt{(3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{4}{3}\right) \approx 53.13^\circ$$

$$z = 5$$

(3) Spherical coordinates:-

A spherical coordinate system is a way to represent points in 3D space using three coordinates:

(1) ρ (rho) (2) θ (3) ϕ

Conversion to Cartesian Coordinates:-

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Conversion from cartesian coordinates:-

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right)$$

Example:- Convert the point $(1, 2, 3)$ in Cartesian coordinates to spherical coordinates.

$$\rho = ? , \theta = ? , \phi = ?$$

$$\text{here } x = 1 , y = 2 , z = 3$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\rho = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1+4+9} = \sqrt{14}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \Rightarrow \tan^{-1}\left(\frac{2}{1}\right) \approx 64.43^\circ$$

$$\phi = \frac{z}{\rho}$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) \Rightarrow \cos^{-1}\left(\frac{3}{\sqrt{14}}\right) \approx 36.70^\circ$$

Example:- Find cartesian coordinates of the point $(4, 45^\circ, 60^\circ)$ in spherical coordinates.

$$\rho = 4, \quad \theta = 45^\circ, \quad \phi = 60^\circ$$

$$x = ?$$

$$y = ?$$

$$z = ?$$

$$x = 4 \sin \phi \cos \theta \Rightarrow 4 \sin(60^\circ) \cos(45^\circ)$$

$$x = 4 \left(\frac{\sqrt{3}}{2}\right) \left(\frac{1}{\sqrt{2}}\right) = \sqrt{6}$$

$$y = 4 \sin \phi \sin \theta = 4 \sin 60^\circ \sin 45^\circ$$

$$y = 4 \left(\frac{\sqrt{3}}{2}\right) \left(\frac{1}{\sqrt{2}}\right) = \sqrt{6}$$

$$z = 4 \cos(60^\circ) = 4 \left(\frac{1}{2}\right) = 2$$

The cartesian points are

$$(x, y, z) = (\sqrt{6}, \sqrt{6}, 2)$$

Derivatives and integrals of vector-valued functions

Given a vector-valued function

$$r(t) = (x(t), y(t), z(t))$$

$$r'(t) = (x'(t), y'(t), z'(t))$$

Example:-

$$r(t) = (t^2, 2t, 3t^2)$$

$$r'(t) = (2t, 2, 6t)$$

Find acceleration of given point

$$r(t) = (t^2, 2t, 3t^2)$$

$$r'(t) = 2t, 2, 6t$$

~~Find~~ $v(t) = 2t, 2, 6t$

$$v(t) = t, 0, 6$$

Integrals of vector-valued functions

$$r(t) = (x(t), y(t), z(t))$$

$$\int r(t) dt = \left(\int x(t) dt, \int y(t) dt, \int z(t) dt \right)$$

Examples:-

$$\int (t^2, 2t, 3t^3) dt$$

$$= \left(\frac{1}{3} t^3, t^2, \frac{3}{4} t^4 \right) + C$$

(2) Find arc length of the curve
 $r(t) = (t^2, 2t, 3t^3)$ from $t = 0$ to $t = 2$

$$\text{Arc length} = \int_{[0, 2]} |r'(t)| dt$$

$$= \int_{[0, 2]} \sqrt{(2t)^2 + 2 + 9t^2} dt$$

$$\text{Arc length} = \int_{[0, 2]} \frac{r'(t)}{dt} (t^2, 2t, 3t^3)$$

$$= \int_{[0, 2]} (2t, 2, 9t) dt$$

Definition:-

The indefinite integral of $\vec{r}$ with respect to t is the set of all anti-derivatives of $\vec{r}$, denoted by $\int \vec{r}(t) dt$. If R is any antiderivative of $\vec{r}$, then

$$\int \vec{r}(t) dt = R(t) + C$$

Examples:-

$$\int ((\cos t)\hat{i} + \hat{j} - 2t\hat{k}) dt$$

Solve:-

$$\int ((\cos t)\hat{i} + \hat{j} - 2t\hat{k}) dt$$

$$= \left(\int \cos t dt \right) \hat{i} + \left(\int dt \right) \hat{j} - \left(\int 2t dt \right) \hat{k}$$

$$= (\sin t + C_1) \hat{i} + (t + C_2) \hat{j} - (t^2 + C_3) \hat{k}$$

$$= \boxed{(\sin t) \hat{i} + t \hat{j} - t^2 \hat{k} + C}$$

where $C = C_1 \hat{i} + C_2 \hat{j} - C_3 \hat{k}$

If the components of $\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$ are integrable over $[a, b]$, then so is $\vec{r}$, and the definite integral of $\vec{r}$ from a to b is

$$\int_a^b \vec{r}(t) dt = \left(\int_a^b f(t) dt \right) \hat{i} + \left(\int_a^b g(t) dt \right) \hat{j} + \left(\int_a^b h(t) dt \right) \hat{k}$$

Example:-

$$\int_0^\pi ((\cos t)\hat{i} + \hat{j} - 2t\hat{k}) dt, \quad [0, \pi]$$

$$\int_0^\pi ((\cos t)\hat{i} + \hat{j} - 2t\hat{k}) dt$$

$$= \left(\int_0^\pi \cos t dt \right) \hat{i} + \left(\int_0^\pi 1 dt \right) \hat{j} - \left(\int_0^\pi 2t dt \right) \hat{k}$$

$$= [\sin t]_0^\pi \hat{i} + [t]_0^\pi \hat{j} - [t^2]_0^\pi \hat{k}$$

$$= [0 - 0] \hat{i} + [\pi - 0] \hat{j} - [\pi^2 - 0^2] \hat{k}$$

$$= \boxed{\pi \hat{j} - \pi^2 \hat{k}} \text{ Ans}$$

Some examples of derivatives and integrals of vector-valued function

$$(1) \int_0^1 [2t^3 \hat{i} + 5\hat{j} + (t-1)\hat{k}] dt$$

$$= \left(\int_0^1 2t^3 dt \right) \hat{i} + \left(\int_0^1 5 dt \right) \hat{j} + \left(\int_0^1 (t-1) dt \right) \hat{k}$$

$$= \left[\frac{2t^4}{4} \right]_0^1 \hat{i} + [5t]_0^1 \hat{j} + \left[\frac{t^2}{2} - t \right]_0^1 \hat{k}$$

$$= \left[\frac{t^4}{2} \right]_0^1 \hat{i} + [5t]_0^1 \hat{j} + \left[\frac{t^2}{2} - t \right]_0^1 \hat{k}$$

$$= \left[\frac{1^4}{2} - \frac{0^4}{2} \right] \hat{i} + [s(1) - s(0)] \hat{j} + [(1)^2 - 0^2 - (1) - 0] \hat{k}$$

$$= \frac{1}{2} \hat{i} + 5 \hat{j} + 0 \hat{k} = \boxed{\frac{1}{2} \hat{i} + 5 \hat{j}}$$

(2) $\int_1^2 \left[(6-6t) \hat{i} + 3\sqrt{t} \hat{j} + \left(\frac{4}{t^2} \right) \hat{k} \right] dt$

$$= \int_1^2 (6-6t) \hat{i} dt + \left(3 \int_1^2 t^{1/2} dt \right) \hat{j} + 4 \left(\int_1^2 t^{-2} dt \right) \hat{k}$$

$$= \left[6t - 3t^2 \right]_1^2 \hat{i} + \left[\frac{8t^{3/2}}{3/2} \right]_1^2 \hat{j} + \left[\frac{4t^{-1}}{-1} \right]_1^2 \hat{k}$$

$$= [6t - 3t^2]_1^2 \hat{i} + [2t^{3/2}]_1^2 \hat{j} + [-4t^{-1}]_1^2 \hat{k}$$

$$= [6(2) - 6(1) - (3(2)^2 - 3(1))] \hat{i} + \left[\frac{8}{3} (2)^{3/2} - \left(\frac{8}{3} \right)^{3/2} \right] \hat{j}$$

$$+ 4 \left[\frac{1}{2} (2)^{-1} - (1)^{-1} \right] \hat{k}$$

$$= -3 \hat{i} + (4\sqrt{2} - 2) \hat{j} + 2 \hat{k} \text{ Ans}$$

3 $\int_0^{\pi/3} [(\sec t \tan t) \hat{i} + (\tan t) \hat{j} + (2 \sin t \cos t) \hat{k}] dt$

$$= \left(\int_0^{\pi/3} (\sec t \tan t) dt \right) \hat{i} + \left(\int_0^{\pi/3} \tan t dt \right) \hat{j} + \left(\int_0^{\pi/3} \sin 2t dt \right) \hat{k}$$

$$= \left[\sec t \right]_0^{\pi/3} \hat{i} + \left[-\ln(\cos t) \right]_0^{\pi/3} \hat{j} + \left[-\frac{1}{2} \cos 2t \right]_0^{\pi/3} \hat{k}$$

$$= \left[\sec(\pi/3) - \sec(0) \right] \hat{i} + \left[-\ln(\cos(\pi/3)) - \cos(0) \right] \hat{j} + \left[-\frac{1}{2} [\cos 2(\pi/3) - \cos 2(0)] \right]$$

$$= \boxed{\hat{i} + (\ln 2) \hat{j} + \frac{3}{4} \hat{k}}$$

⇒ Arc Length in space

The length of a smooth curve $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, $a \leq t \leq b$, that is traced exactly once as t increases from $t = a$ to $t = b$ is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \quad \text{--- (1)}$$

As $|V| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$

So

$$L = \int_a^b |V| dt \quad \text{--- (2)}$$

Example:- A glider is soaring upward along the helix $\vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j} + t\hat{k}$. How long is the glider's path from $t = 0$ to $t = 2\pi$?

$$L = \int_a^b |V| dt = \int_0^{2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2} dt$$

$$= \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + 1} dt$$

$$\int_0^{2\pi} \sqrt{2} dt = \boxed{2\pi\sqrt{2}} \quad \text{Ans units length}$$

=> Arc length Parameter with
base point $P(t_0)$

$$s(t) = \int_{t_0}^t \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2} d\tau$$
$$= \int_{t_0}^t |v(\tau)| d\tau \quad \text{--- (3)}$$

Example

$$\vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j} + t\hat{k}$$

For $v(t)$ we differentiate $\vec{r}(t)$

$$\frac{d\vec{r}(t)}{dt} = (\sin t)\hat{i} + (\cos t)\hat{j} + \hat{k}$$

$$v(\tau) = (-\sin \tau)\hat{i} + (\cos \tau)\hat{j} + \hat{k}$$

$$|v(\tau)| = \sqrt{(-\sin \tau)^2 + (\cos \tau)^2 + (1)^2}$$

$$|v(\tau)| = \sqrt{2}$$

Now

$$s(t) = \int_0^t \sqrt{2} d(\tau)$$

$$\boxed{t = \frac{s}{\sqrt{2}}}$$

Then

$$\vec{r}(t) = \left(\cos \frac{s}{\sqrt{2}}\right)\hat{i} + \left(\sin \frac{s}{\sqrt{2}}\right)\hat{j} + \frac{s}{\sqrt{2}}\hat{k}$$

Unit tangent vector

$$\vec{T} = \frac{\vec{v}}{|\vec{v}|}$$

★ Finding Tangent vectors and lengths

$$\textcircled{1} \quad \vec{r}(t) = (1 + 2\cos t)\hat{i} + (2\sin t)\hat{j} + \sqrt{5}t\hat{k}, \quad 0 \leq t \leq \pi$$

$$\vec{v}(t) = \frac{d}{dt}(1 + 2\cos t)\hat{i} + \frac{d}{dt}(2\sin t)\hat{j} + \frac{d}{dt}(\sqrt{5}t)\hat{k}$$

$$\vec{v}(t) = (-2\sin t)\hat{i} + (2\cos t)\hat{j} + \sqrt{5}\hat{k}$$

$$|\vec{v}| = \sqrt{(-2\sin t)^2 + (2\cos t)^2 + (\sqrt{5})^2}$$

$$|\vec{v}| = \sqrt{4\sin^2 t + 4\cos^2 t + 5}$$

$$|\vec{v}| = \sqrt{4(\sin^2 t + \cos^2 t) + 5} = \sqrt{4 + 5} = 3$$

$$\text{Tangent vector} = \frac{\vec{v}}{|\vec{v}|} = \frac{(-2\sin t)\hat{i} + (2\cos t)\hat{j} + \sqrt{5}\hat{k}}{3}$$

$$= \left[\left(-\frac{2}{3}\sin t\right)\hat{i} + \left(\frac{2}{3}\cos t\right)\hat{j} + \frac{\sqrt{5}}{3}\hat{k} \right]$$

$$\text{Length} = \int_a^b |\vec{v}| dt = \int_0^\pi 3 dt = [3t]_0^\pi$$

$$= \boxed{3\pi} \text{ Ans.}$$

$$\textcircled{2} \quad r(t) = 2t\hat{i} + \frac{2}{5}t^{3/2}\hat{k}, \quad 0 \leq t \leq 5$$

$$v(t) = 2\hat{i} + \left(\frac{2}{5} \times \frac{3}{2}\right)t^{1/2}\hat{k}$$

$$v(t) = 2\hat{i} + \frac{3}{5}t^{1/2}\hat{k}$$

$$|v(t)| = \sqrt{(2)^2 + \left(\frac{3}{5}t^{1/2}\right)^2} = \sqrt{4 + \frac{9}{25}t}$$

$$\text{Vector} = \frac{v}{|v|} = \frac{2\hat{i} + \frac{3}{5}t^{1/2}\hat{k}}{\sqrt{4 + \frac{9t}{25}}} \quad \checkmark$$

$$\text{Arc length} = \int_0^5 \sqrt{4 + \frac{9t}{25}} dt$$

$$= \left[\frac{\left(4 + \frac{9t}{25}\right)^{3/2}}{3/2} \right]_0^5 = \left[\frac{2 \left(4 + \frac{9t}{25}\right)^{3/2}}{3} \right]_0^5$$

$$= \frac{2 \left(4 + \frac{9(5)}{25}\right)^{3/2}}{3} = \frac{2 \left(\frac{100 + 45}{25}\right)^{3/2}}{3}$$

$$= \frac{2 \left(\frac{145}{25}\right)^{3/2}}{3} \quad \checkmark$$

$$(3) \quad r(t) = (\cos^3 t) \hat{j} + (\sin^3 t) \hat{k}, \quad 0 \leq t \leq \pi/2$$

$$v(t) = (-3\cos^2 t \sin t) \hat{j} + (3\sin^2 t \cos t) \hat{k}$$

$$|v| = \sqrt{(-3\cos^2 t \sin t)^2 + (3\sin^2 t \cos t)^2}$$

$$= \sqrt{(9\cos^2 t \sin^2 t)(\cos^2 t + \sin^2 t)}$$

$$= 3|\cos t \sin t|$$

$$\hat{T} = \frac{v}{|v|} = \frac{(-3\cos^2 t \sin t) \hat{j} + (3\sin^2 t \cos t) \hat{k}}{3|\cos t \sin t|}$$

the length

$$L = \int_0^{\pi/2} 3|\cos t \sin t| dt$$

$$= \int_0^{\pi/2} 3 \cos t \sin t dt$$

$$= \int_0^{\pi/2} \frac{3}{2} \sin 2t dt$$

$$= \left[-\frac{3}{4} \cos 2t \right]_0^{\pi/2} = \left[\frac{3}{2} \right]$$

For Finding Arc length Parameter

$$(1) \quad r(t) = (\cos t + t \sin t) \hat{i} + (\sin t - t \cos t) \hat{j}$$

$$, \quad \pi/2 \leq t \leq \pi$$

$$r(t) = (\cos t + t \sin t) \hat{i} + (\sin t - t \cos t) \hat{j}$$

$$v(t) = (-\sin t + \sin t + t \cos t) \hat{i} + (\cos t - \cos t + t \sin t) \hat{j}$$

$$= (t \cos t) \hat{i} + (t \sin t) \hat{j}$$

$$|v| = \sqrt{(t \cos t)^2 + (t \sin t)^2} = \sqrt{t^2} = t$$

As we know

$$s(t) = \int_{\pi/2}^{\pi} r dr = \int_{\pi/2}^{\pi} t dr$$

$$s(t) = \left[\frac{t^2}{2} \right]_{\pi/2}^{\pi}$$

$$= \frac{\pi^2}{2} - \frac{\left(\frac{\pi}{2}\right)^2}{2}$$

$$= \frac{\pi^2}{2} - \frac{\frac{\pi^2}{4}}{2}$$

$$= \frac{\pi^2 - \frac{\pi^2}{4}}{2}$$

$$= \frac{\frac{4\pi^2 - \pi^2}{4}}{2} = \frac{4\pi^2 - \pi^2}{8}$$

$$= \frac{3\pi^2}{8} \text{ Ans}$$

② $r(t) = (1-3t)\hat{i} + (1-2t)\hat{j} + 6t\hat{k}, 0 \leq t \leq 1$

$$v(t) = \frac{d}{dt} (1-3t)\hat{i} + \frac{d}{dt} (1-2t)\hat{j} + \frac{d}{dt} (6t)\hat{k}$$

$$v(t) = -3\hat{i} - 2\hat{j} + 6\hat{k}$$

$$|v| = \sqrt{(-3)^2 + (-2)^2 + (6)^2} = \sqrt{9+4+36}$$

$$|v| = \sqrt{49} = 7$$

Formula of parameter arc length

$$s(t) = \int_0^1 |v| r dr$$

$$s(t) = \int_0^t 7 \, d\tau$$

$$s(t) = [7t]_0^t$$

$$s(t) = (7t)$$

Curvature of a plane curve, normal and binormal vectors

Curvature

The traditional symbol for the curvature function is the Greek letter κ ("Kappa")

If $\mathbf{T}$ is the unit vector of a smooth curve, the curvature function of the curve is

$$\kappa = \left| \frac{d\mathbf{T}}{ds} \right| \quad \text{--- (1)}$$

$$= \left| \frac{d\mathbf{T}}{ds} \frac{dt}{ds} \right| = \frac{1}{|ds/dt|} \left| \frac{d\mathbf{T}}{dt} \right|$$

$$\kappa = \frac{1}{|V|} \left| \frac{d\mathbf{T}}{dt} \right| \quad \text{--- (2)}$$

Example:-

$$r(t) = (a \cos t) \hat{i} + (a \sin t) \hat{j}$$

$$v = \frac{dr}{dt} = -a \sin t \hat{i} + a \cos t \hat{j}$$

$$|v| = \sqrt{(-a \sin t)^2 + (a \cos t)^2} = \sqrt{a^2} = a$$

$$\hat{T} = \frac{v}{|v|} = \frac{(-a \sin t) \hat{i} + (a \cos t) \hat{j}}{a}$$

$$\hat{T} = \frac{(-a \sin t) \hat{i} + (a \cos t) \hat{j}}{a} = -\sin t \hat{i} + \cos t \hat{j}$$

$$\frac{d\hat{T}}{dt} = -(\cos t) \hat{i} - (\sin t) \hat{j}$$

$$\left| \frac{d\hat{T}}{dt} \right| = \sqrt{\cos^2 t + \sin^2 t} = 1$$

Hence

$$K = \frac{1}{|v|} \left| \frac{d\hat{T}}{dt} \right| = \frac{1}{a} (1) = \boxed{\frac{1}{a}}$$

⊛ At a point where $K \neq 0$ the principal unit normal vector for a smooth curve in the plane is

$$N = \frac{1}{K} \frac{d\hat{T}}{ds}$$

Formula of calculating N

is

$$N = \frac{d\hat{T}/dt}{|d\hat{T}/dt|}$$

where $\hat{T}$ is tangential vector
vector $\hat{T} = \frac{v}{|v|}$

Example:-

$$r(t) = (\cos 2t)\hat{i} + (\sin 2t)\hat{j}$$

$$T = ? , N = ?$$

First Find T :

$$r(t) = (\cos 2t)\hat{i} + (\sin 2t)\hat{j}$$

$$v = (-2\sin 2t)\hat{i} + (2\cos 2t)\hat{j}$$

$$|v| = \sqrt{(-2\sin 2t)^2 + (2\cos 2t)^2}$$

$$= \sqrt{4\sin^2 2t + 4\cos^2 2t} = \sqrt{4(\sin^2 2t + \cos^2 2t)}$$

$$|v| = \sqrt{4(1)} = 2$$

$$T = \frac{v}{|v|} = \frac{(-2\sin 2t)\hat{i} + (2\cos 2t)\hat{j}}{2}$$

$$T = (-\sin 2t)\hat{i} + (\cos 2t)\hat{j}$$

For $N = ?$

$$\frac{dT}{dt} = (-2\cos 2t)\hat{i} - (2\sin 2t)\hat{j}$$

$$= -2\cos 2t\hat{i} - 2\sin 2t\hat{j}$$

$$\left| \frac{dT}{dt} \right| = \sqrt{(-2\cos 2t)^2 + (-2\sin 2t)^2}$$

$$= \sqrt{4\cos^2 2t + 4\sin^2 2t}$$

$$= \sqrt{4(\cos^2 2t + \sin^2 2t)}$$

$$= \sqrt{4(1)} = 2$$

$$\text{Now, } N = \frac{d\hat{T}/dt}{|d\hat{T}/dt|} = \frac{-2\cos 2t\hat{i} - 2\sin 2t\hat{j}}{2}$$

$$N = (-\cos 2t)\hat{i} - (\sin 2t)\hat{j}$$

Ⓐ Questions from 13.4 in 14th edition

① $r(t) = t\hat{i} + \ln(\sin t)\hat{j}$, $-\pi < t < \pi$

$\hat{T} = ?$, $N = ?$, $K = ?$

For $\hat{T}$:

$$r(t) = t\hat{i} + \ln(\sin t)\hat{j}$$

$$v = \hat{i} + \left(\frac{1}{\sin t} \cos t\right)\hat{j}$$

$$v = \hat{i} + \left(\frac{\cos t}{\sin t}\right)\hat{j}$$

$$v = \hat{i} + (\cot t)\hat{j}$$

$$|v| = \sqrt{(1)^2 + (\cot t)^2}$$

$$|v| = \sqrt{1 + \cot^2 t}$$

$$\hat{T} = \frac{v}{|v|} = \frac{\hat{i} + (\cot t)\hat{j}}{\sqrt{1 + \cot^2 t}}$$

For N :

$$N = \frac{d\hat{T}/dt}{|d\hat{T}/dt|} = \dots$$

For $K =$

$$K = \frac{1}{|v|} \left| \frac{d\hat{T}}{dt} \right| = \dots$$

$$\textcircled{2} \quad \mathbf{r}(t) = \ln(\cos t) \hat{i} + t \hat{j}, \quad -\pi < t < \pi$$

$$K = ?, \quad T = ?, \quad N = ?$$

$$T = ?$$

$$\mathbf{r}(t) = \ln(\cos t) \hat{i} + t \hat{j}$$

$$\mathbf{v} = \left(\frac{-\sin t}{\cos t} \right) \hat{i} + \hat{j}$$

$$\mathbf{v} = (-\tan t) \hat{i} + \hat{j}$$

$$|\mathbf{v}| = \sqrt{(-\tan t)^2 + (1)^2}$$

$$|\mathbf{v}| = \sqrt{\tan^2 t + 1} = \sqrt{\sec^2 t} = \sec t$$

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{(-\tan t) \hat{i} + \hat{j}}{\sec t}$$

~~$$\mathbf{T} = \frac{(-\tan t) \hat{i} + \hat{j}}{\sec t}$$~~

$$\mathbf{T} = \left(\frac{-\tan t}{\sec t} \right) \hat{i} + \left(\frac{1}{\sec t} \right) \hat{j}$$

$$\boxed{\mathbf{T} = (-\sin t) \hat{i} + (\cos t) \hat{j}} \quad \checkmark$$

$$\frac{d\mathbf{T}}{dt} = (-\cos t) \hat{i} - (\sin t) \hat{j}$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \sqrt{(-\cos t)^2 + (-\sin t)^2}$$

$$= \sqrt{(\cos^2 t + \sin^2 t)} = \sqrt{1} = 1$$

$$\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|} = \frac{(-\cos t) \hat{i} - (\sin t) \hat{j}}{1}$$

$$\boxed{\mathbf{N} = (-\cos t) \hat{i} - (\sin t) \hat{j}} \quad \checkmark$$

$$K = \frac{1}{|v|} \left| \frac{d\hat{T}}{dt} \right| = \frac{1}{\sec t} (1)$$

$$K = \frac{1}{\sec t} = \boxed{\cos t} \checkmark$$

③ $r(t) = (2t+3)\hat{i} + (5-t^2)\hat{j}$

Now, K and $\hat{T}$ Find.

$$r(t) = (2t+3)\hat{i} + (5-t^2)\hat{j}$$

$$v = (2)\hat{i} - (2t)\hat{j}$$

$$|v| = \sqrt{(2)^2 + (-2t)^2} = \sqrt{4 + 4t^2}$$

$$= \sqrt{4(1+t^2)}$$

$$|v| = 2\sqrt{1+t^2}$$

Now

$$\hat{T} = \frac{v}{|v|} = \frac{2\hat{i} - (2t)\hat{j}}{2\sqrt{1+t^2}}$$

$$\hat{T} = \left[\frac{\hat{i}}{\sqrt{1+t^2}} - \frac{t}{\sqrt{1+t^2}} \hat{j} \right] \checkmark$$

$$\frac{d\hat{T}}{dt} = \frac{-t}{(\sqrt{1+t^2})^3} \hat{i} - \frac{1}{(\sqrt{1+t^2})^3} \hat{j}$$

$$\left| \frac{d\hat{T}}{dt} \right| = \sqrt{\left(\frac{-t}{(\sqrt{1+t^2})^3} \right)^2 + \left(\frac{-1}{(\sqrt{1+t^2})^3} \right)^2}$$

$$\left| \frac{d\hat{T}}{dt} \right| = \sqrt{\frac{1}{(1+t^2)^2}} = \frac{1}{1+t^2}$$

$$N = \frac{d\pi'/dt}{|d\pi'/dt|} = \frac{-t \hat{i} - \frac{1}{\sqrt{1+t^2}} \hat{j}}{\left(\sqrt{1+t^2}\right) \left(\sqrt{1+t^2}\right)^{3/2}}$$

$$N = \left[\frac{-t}{\sqrt{1+t^2}} \hat{i} - \frac{1}{\sqrt{1+t^2}} \hat{j} \right] \checkmark$$

$$K = \frac{1}{|v|} \left| \frac{d\pi'}{dt} \right|$$

$$= \frac{1}{2\sqrt{1+t^2}} \left(\frac{1}{1+t^2} \right) = \boxed{\frac{1}{2(1+t^2)^{3/2}}} \checkmark$$

Binomial vectors

$$B = \pi' \times N$$

Dot products

$$\textcircled{1} \Rightarrow \vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 \quad \textcircled{1}$$

$$\Rightarrow \boxed{\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta} \quad \textcircled{2}$$

$$\Rightarrow \boxed{\theta = \cos^{-1} \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}}$$

Two vectors are orthogonal vectors if their dot product is zero i.e.

$$\vec{u} \cdot \vec{v} = 0$$

② Properties:-

$$1. \quad \vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$$

$$2. \quad (c\vec{u}) \cdot \vec{v} = \vec{u} \cdot (c\vec{v}) = c(\vec{u} \cdot \vec{v})$$

$$3. \quad \vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$$

$$4. \quad \vec{u} \cdot \vec{u} = |\vec{u}|^2$$

$$5. \quad 0 \cdot \vec{u} = 0$$

③ Vector projection of $\vec{u}$ on $\vec{v}$ is the vector

$$\text{Proj}_{\vec{v}} \vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v} = \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} \right) \left(\frac{\vec{v}}{|\vec{v}|} \right)$$

④ The scalar component of u in the direction of v is the scalar

$$\boxed{|\vec{u}| \cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} = \vec{u} \cdot \frac{\vec{v}}{|\vec{v}|}}$$

Questions:

① $\left(\frac{1}{2}\hat{i} + 3\hat{j} + \hat{k}\right) \cdot (4\hat{i} - \hat{j} + 2\hat{k}) = ?$

$\therefore 1 = \left(\frac{1}{2}\right)(4) + (3)(-1) + (1)(2)$

$\therefore 1 = \boxed{1}$

② $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}$ and $\vec{v} = 6\hat{i} + 3\hat{j} + 2\hat{k}$
 $\theta = ?$

$$\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \quad \text{--- (1)}$$

Now

$$\vec{u} \cdot \vec{v} = (1)(6) + (-1)(3) + (-2)(2)$$

$$= 6 - 3 - 4 = -1$$

$$|\vec{u}| = \sqrt{(1)^2 + (-1)^2 + (-2)^2} = \sqrt{1 + 1 + 4}$$

$$= \sqrt{6} = \sqrt{6} = \sqrt{6}$$

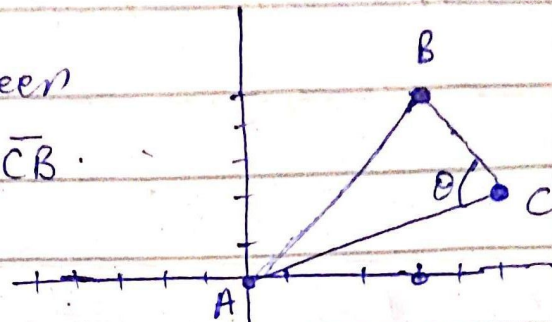
$$|\vec{v}| = \sqrt{(6)^2 + (3)^2 + (2)^2} = \sqrt{36 + 9 + 4} = \sqrt{49} = 7$$

Put in eq (1)

$$\theta = \frac{-1}{(\sqrt{6})(7)} = \frac{-1}{7\sqrt{6}} \approx 1.76 \text{ rad}$$

③ Find the θ in the triangle ABC determined by vertices $A = (0,0)$, $B = (3,5)$, and $C = (5,2)$

The angle is between the vector $\vec{CA}$ and $\vec{CB}$.



$$\vec{CA} = (-5, -2), \quad \vec{CB} = (-2, 3)$$

Now

$$\theta = \cos^{-1} \frac{\vec{CA} \cdot \vec{CB}}{|\vec{CA}| |\vec{CB}|}$$

$$\vec{CA} \cdot \vec{CB} = (-5)(-2) + (-2)(3) = 10 - 6 = 4$$

$$|\vec{CA}| = \sqrt{(-5)^2 + (-2)^2} = \sqrt{25 + 4} = \sqrt{29}$$

$$|\vec{CB}| = \sqrt{(-2)^2 + (3)^2} = \sqrt{4 + 9} = \sqrt{13}$$

$$\theta = \cos^{-1} \frac{4}{\sqrt{29} \sqrt{13}} = 78.1^\circ \text{ or } 1.36 \text{ radians}$$

④ Determine these vectors are orthogonal.

(a) $\vec{u} = (3, -2)$ and $\vec{v} = (4, 6)$

Solve:- $\vec{u} \cdot \vec{v} = (3)(4) + (-2)(6)$

$$\vec{u} \cdot \vec{v} = 12 - 12 = 0$$

$\boxed{\vec{u} \cdot \vec{v} = 0}$ Therefore orthogonal.

$$(b) \vec{u} = 3\hat{i} - 2\hat{j} + \hat{k} \text{ and } \vec{v} = 2\hat{j} + 4\hat{k}$$

$$\vec{u} \cdot \vec{v} = (3)(0) + (-2)(2) + (1)(4)$$

$$= 0 - 4 + 4 = 0$$

These are orthogonal vectors.

(5) Find the vector projection of $\vec{u} = 6\hat{i} + 3\hat{j} + 2\hat{k}$ onto $\vec{v} = \hat{i} - 2\hat{j} - 2\hat{k}$ and also the scalar component of $\vec{u}$ in the direction of $\vec{v}$.

$$\text{Proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v} = \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v}$$

$$= \frac{6 - 6 - 4}{(3)^2} (\hat{i} - 2\hat{j} - 2\hat{k})$$

$$= \frac{-4}{9} (\hat{i} - 2\hat{j} - 2\hat{k})$$

$$= \left[-\frac{4}{9}\hat{i} + \frac{8}{9}\hat{j} + \frac{8}{9}\hat{k} \right]$$

We find the scalar component of $\vec{u}$ in the direction of $\vec{v}$ ~~from~~

$$|\vec{u}| \cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|}$$

$$= \frac{6\hat{i} + 3\hat{j} + 2\hat{k} \cdot (\hat{i} - 2\hat{j} - 2\hat{k})}{3}$$

$$= \frac{6 - 6 - 4}{3}$$

$$= \left(-\frac{4}{3} \right)$$

⑥ Find the vector projection of a force $\vec{F} = 5\hat{i} + 2\hat{j}$ onto $\vec{v} = \hat{i} - 3\hat{j}$ and scalar component of $\vec{F}$ in the direction of $\vec{v}$.

$$\begin{aligned}\text{Proj}_{\vec{v}} \vec{F} &= \frac{\vec{F} \cdot \vec{v}}{|\vec{v}|^2} \vec{v} \\ &= \frac{(5) - 6}{10} (\hat{i} - 3\hat{j}) \\ &= \frac{-1}{10} (\hat{i} - 3\hat{j})\end{aligned}$$

$$\text{Proj}_{\vec{v}} \vec{F} = \left[-\frac{1}{10} \hat{i} + \frac{3}{10} \hat{j} \right]$$

Scalar component of $\vec{F}$ in the direction of $\vec{v}$

$$= \frac{\vec{F} \cdot \vec{v}}{|\vec{v}|}$$

$$= \frac{5 - 6}{\sqrt{10}} = \left[\frac{-1}{\sqrt{10}} \right]$$

⑦ If $|\vec{F}| = 40\text{N}$, $|\vec{D}| = 3\text{m}$ and $\theta = 60^\circ$, the work done by $\vec{F}$ in acting from P to Q is

$$\text{Work} = \vec{F} \cdot \vec{D}$$

$$W = F|\vec{D}| \cos \theta$$

$$W = (40)(3) \cos 60^\circ = 60\text{J}$$

Exercise 12.3.

① $\vec{v} = \hat{i} - 2\hat{j} + \sqrt{3}\hat{k}$, $\vec{u} = -\hat{i} + 2\hat{j} - \sqrt{3}\hat{k}$

(a) $\vec{v} \cdot \vec{u} = (-1 - 4 - 3) = -8$

$$|\vec{v}| = \sqrt{1+4+3} = \sqrt{8}$$

$$|\vec{u}| = \sqrt{1+4+3} = \sqrt{8}$$

(b) $\cos \theta = \frac{\vec{v} \cdot \vec{u}}{|\vec{v}| |\vec{u}|} = \frac{-8}{(\sqrt{8})(\sqrt{8})} = \frac{-8}{8} = -1$

(c) $|\vec{u}| \cos \theta = \frac{\vec{v} \cdot \vec{u}}{|\vec{v}|} = \frac{-8}{\sqrt{8}} = -\sqrt{8}$

(d) The vector projection of $\sqrt{2}$ u onto v or $\text{Proj}_v u$

$$= \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v} = \left(\frac{-8}{8} \right) \hat{i} - 2\hat{j} + \sqrt{3}\hat{k}$$

$$= -\hat{i} + 2\hat{j} - \sqrt{3}\hat{k}$$

④ $\vec{v} = 2\hat{i} + 10\hat{j} - 11\hat{k}$, $\vec{u} = 2\hat{i} + 2\hat{j} + \hat{k}$

$$\vec{v} \cdot \vec{u} = (4 + 20 - 11) = 13$$

$$|\vec{v}| = \sqrt{4 + 100 + 121} = \sqrt{225} = 15$$

$$|x| = \sqrt{4+4+1} = \sqrt{9} = 3$$

$$(b) \cos \theta = \frac{\vec{v} \cdot \vec{u}}{|\vec{v}| |\vec{u}|} = \frac{13}{(15)(3)} = \boxed{\frac{13}{45}}$$

(c) $\vec{u} \cos \theta = \frac{\vec{v} \cdot \vec{u}}{|\vec{v}|} = \frac{13}{15}$

$$(d) \quad \text{Proj}_{\vec{v}} \vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v} = \left(\frac{13}{225} \right) 2\hat{i} + 10\hat{j} - 11\hat{k}$$

$$= \frac{26}{225} \hat{i} + \frac{26}{45} \hat{j} - \frac{11}{45} \hat{k}$$

$$(7) \vec{v} = 2\hat{i} + \hat{j}, \vec{u} = \sqrt{2}\hat{i} + \sqrt{15}\hat{j}$$

$$(a) \vec{v} \cdot \vec{u} = (2\sqrt{2} + \sqrt{15})$$

$$|\vec{v}| = \sqrt{4+1} = \sqrt{5}$$

$$|\vec{u}| = \sqrt{2+15} = \sqrt{17}$$

$$(b) \cos \theta = \frac{\vec{v} \cdot \vec{u}}{|\vec{v}| |\vec{u}|} = \frac{2\sqrt{2} + \sqrt{15}}{(\sqrt{5})(\sqrt{17})} =$$

$$(c) |\vec{u}| \cos \theta = \frac{\vec{v} \cdot \vec{u}}{|\vec{v}|} = \frac{2\sqrt{2} + \sqrt{15}}{\sqrt{5}}$$

$$(d) \text{Proj}_{\vec{u}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{u}}{|\vec{u}|^2} \right) \vec{u}$$

$$= \left(\frac{2\sqrt{2} + \sqrt{15}}{5} \right) (2\hat{i} + \hat{j})$$

Find angle.

$$(9) \vec{u} = \hat{i} + 2\hat{j}, \vec{v} = 2\hat{j} + \hat{j} + \hat{k}$$

$$\theta = \cos^{-1} \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \right)$$

$$\theta = \cos^{-1} \left(\frac{2+2}{(\sqrt{5})(\sqrt{6})} \right) = \cos^{-1} \left(\frac{4}{\sqrt{5}\sqrt{6}} \right)$$

$$\theta = 0.75 \text{ radians.}$$

$$(12) \vec{u} = \sqrt{3}\hat{i} - \sqrt{3}\hat{k}, \vec{v} = -3\hat{i} + 3\hat{j} - 3\hat{k}$$

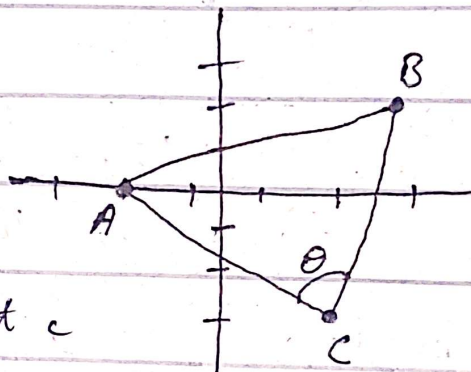
$$\theta = \cos^{-1} \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \right) = \frac{-3\sqrt{3} + 3\sqrt{3}}{(\sqrt{6})(\sqrt{27})}$$

$$\theta = \cos^{-1} \left(\frac{0}{\sqrt{6} \sqrt{27}} \right) = \cos^{-1}(0) = \frac{\pi}{2}$$

1.57 radians $\theta = \frac{\pi}{2} = 1.57 \text{ radians}$

(13)

$$A = (-2, 0), B(3, 2), C = (2, -3)$$



let angle at c

$$\vec{u} = \vec{CA} = -4\hat{i} + 3\hat{j}$$

$$\vec{v} = \vec{CB} = \hat{i} + 5\hat{j}$$

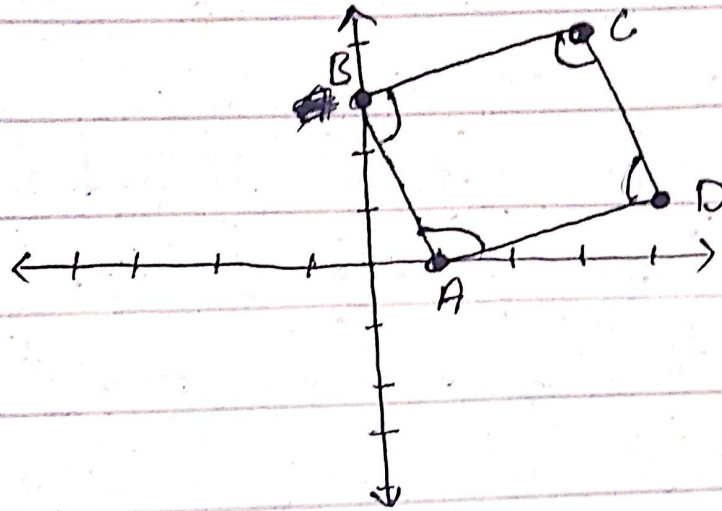
$$\theta = \cos^{-1} \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \right) = \cos^{-1} \left(\frac{-4 + 15}{\sqrt{25} \sqrt{26}} \right)$$

$$\theta = \cos^{-1} \left(\frac{11}{5\sqrt{26}} \right)$$

$$\theta = 1.124 \text{ radian at } C$$

we can find angle at a and b.

(14) $A = (1, 0)$, $B = (0, 3)$, $C = (3, 4)$, $D = (4, 1)$



$$\vec{u} = \overrightarrow{AC} = 2\hat{i} + 4\hat{j}$$

$$\vec{v} = \overrightarrow{BD} = 4\hat{i} - 2\hat{j}$$

$$\theta = \cos^{-1} \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|} \right) = \cos^{-1} \left(\frac{8 - 8}{(\sqrt{20})(\sqrt{20})} \right) = \cos^{-1} \frac{0}{20}$$

$$\theta = \pi$$

$$\theta = 1.77 \text{ radians}$$

* Cross product

* The Cross product

$$\vec{u} \times \vec{v} = (|\vec{u}| |\vec{v}| \sin \theta) \hat{n}$$

Parallel vectors.

Nonzero vectors $\vec{u}$ and $\vec{v}$ are parallel if and only if $\vec{u} \times \vec{v} = 0$

Properties of cross product

If $\vec{u}, \vec{v}$ and $\vec{w}$ are any vectors and r, s are scalars, then

$$(1) (r\vec{u}) \times (s\vec{v}) = (rs)(\vec{u} \times \vec{v})$$

$$(2) \vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w} \quad (3) \vec{v} \times \vec{u} = -(\vec{u} \times \vec{v})$$

$$(4) (\vec{v} + \vec{w}) \times \vec{u} = \vec{v} \times \vec{u} + \vec{w} \times \vec{u}$$

$$(5) 0 \times \vec{u} = 0 \quad (6) \vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w})\vec{v} - (\vec{u} \cdot \vec{v})\vec{w}$$

$$\Rightarrow \text{Area of Parallelogram} = |\vec{u} \times \vec{v}|$$

$$\Rightarrow \text{Area of triangle} = \frac{1}{2} |\vec{u} \times \vec{v}|$$

Ex $\vec{u} = 2\hat{i} + \hat{j} + \hat{k}, \vec{v} = -4\hat{i} + 3\hat{j} + \hat{k}$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix} = \hat{i}(1-3) - \hat{j}(2+4) + \hat{k}(6+4)$$
$$= -2\hat{i} - 6\hat{j} + 10\hat{k}$$

$$\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u}) = 2\hat{i} + 6\hat{j} - 10\hat{k}$$

Ex 2 $P(1, -1, 0), Q(2, 1, -1), R(-1, 1, 2)$

$$\vec{v} = \vec{PQ} = \hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{u} = \vec{PR} = -2\hat{j} + 2\hat{j} + 2\hat{k}$$

$$\vec{v} \times \vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$= (4+2)\hat{i} - (2-2)\hat{j} + (2+4)\hat{k}$$

$$\vec{v} \times \vec{u} = \boxed{6\hat{i} + 6\hat{k}} \text{ Ans}$$

Ex 3 $P(1, -1, 0), Q(2, 1, -1)$
 $R(-1, 1, 2)$

As we know area of triangle is

$$= \frac{1}{2} |\vec{v} \times \vec{u}|$$

$$\vec{v} = \vec{PQ} = \hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{u} = \vec{PR} = -2\hat{j} + 2\hat{j} + 2\hat{k}$$

$$= \frac{1}{2} |\vec{v} \times \vec{u}| = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = \frac{1}{2} (6\hat{i} + 6\hat{k})$$

$$\begin{array}{r} 6 \overline{) 72} \\ 6 \overline{) 12} \\ \hline 6 \sqrt{2} \end{array}$$

$$\frac{1}{2} \sqrt{6^2 + 6^2}$$

$$\frac{1}{2} \sqrt{36 + 36} = \sqrt{72}$$

$$\frac{1}{2} 6\sqrt{2}$$

$$3\sqrt{2}$$

Ex $P(0,0,0), Q(1,2,3), R(-1,2,0)$

$$\vec{v} = \overrightarrow{PQ} = \hat{i} + 2\hat{j} + 3\hat{k}, \quad \vec{u} = \overrightarrow{PR} = -\hat{i} + 2\hat{j}$$

$$\vec{v} \times \vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ -1 & 2 & 0 \end{vmatrix} = -6\hat{i} + 3\hat{j} + 4\hat{k}$$

$$|\vec{v} \times \vec{u}| = \sqrt{36 + 9 + 16} = \sqrt{61}$$

As we know area of triangle is

$$= \frac{1}{2} |\vec{v} \times \vec{u}|$$

$$= \frac{1}{2} \sqrt{61}$$

[Ex 4] $P(1,-1,0), Q(2,-1,-1), R(-1,1,2)$

$$\vec{v} = \overrightarrow{PQ} = \hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{u} = \overrightarrow{PR} = -2\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{v} \times \vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$\vec{v} \times \vec{u} = 6\hat{i} + 6\hat{k}, \quad |\vec{v} \times \vec{u}| = 6\sqrt{2}$$

$$\frac{\vec{v} \times \vec{u}}{|\vec{v} \times \vec{u}|} = \frac{6\hat{i} + 6\hat{k}}{6\sqrt{2}} \text{ Ans.}$$

Scalar triple product

$$\overline{(\vec{u} \times \vec{v})} \cdot \vec{w} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

It is also called volume of parallelepiped.

Ex 6

$$\vec{u} = \hat{i} + 2\hat{j} - \hat{k} \quad \vec{v} = -2\hat{i} + 3\hat{k}$$

$$\vec{w} = 7\hat{j} - 4\hat{k}$$

$$\overline{(\vec{u} \times \vec{v})} \cdot \vec{w} = \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix} = 1(-21) - 2(-14) - 1(-14)$$
$$= -21 + 16 + 14$$
$$= -21 + 30 = 9$$

$$= 1(-21) - 2(-14) + (-1)(14)$$

$$= -21 + 16 - 14$$

$$= -35 + 16 = -19$$

The volume

$$\text{is } |\overline{(\vec{u} \times \vec{v})} \cdot \vec{w}| = 19$$

units cubed

Show $\bar{u} - \text{Proj}_v u \perp \text{Proj}_v u$.

Solve

Because $\text{Proj}_v u$ is parallel to $\vec{v}$. It is enough to show $\bar{u} - \text{Proj}_v u \perp \text{Proj}_v u$.

So take

$$(\bar{u} - \text{Proj}_v u) \cdot (\bar{v})(\bar{v}) = \bar{u} \cdot \bar{v} - \text{Proj}_v u \cdot \bar{v}$$

$$\begin{aligned} \text{As Proj} &= \bar{u} \cdot \bar{v} - \frac{\bar{u} \cdot \bar{v}}{\bar{v} \cdot \bar{v}} \bar{v} \cdot \bar{v} \\ &= \bar{u} \cdot \bar{v} - \bar{u} \cdot \bar{v} \end{aligned}$$

$$= 0$$

So

$$\bar{u} - \text{Proj}_v u \perp \text{Proj}_v u$$

12.5



Lines and Planes in space

⊛ Vector equation of a line

A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to $\vec{v}$ is

$\vec{r}(t) = \vec{r}_0 + \vec{v}t$

↑ Fixed point ↑ Parallel vector
parameter

Parametric eqs

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}$$

Ex Find parametric eq of line passes through $P(-3, 2, -3)$ and $Q(1, -1, 4)$

$$\vec{v} = \vec{PQ} = 4\hat{i} - 3\hat{j} + 7\hat{k}$$

Let $(-3, 2, -3)$ be the point through which line L passes.

$$\text{So } x = x_0 + at = -3 + 4t$$

$$y = y_0 + bt = 2 + (-3t) = 2 - 3t$$

$$z = z_0 + ct = -3 + 7t$$

Example 1

$$P(-2, 0, 4), \vec{v} = 2\hat{i} + 4\hat{j} - 2\hat{k}$$

$$x = x_0 + at$$

$$x = -2 + 2t$$

$$y = y_0 + bt = 4t$$

$$z = z_0 + ct = 4 + (-2)t = 4 - 2t$$

Example 4

(1, 1, 1)

$$\text{unit vector } \hat{v} = \hat{u} = \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}$$

we know position vector

$$r(t) = r_0 + t\vec{v}$$

$$\text{for } \vec{v} = |\vec{v}| \hat{v}$$

$$r(t) = 0 + t(|\vec{v}| \hat{v})$$

$$r(t) = 0 + t(60) \left(\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k} \right)$$

$$r(t) = 20\sqrt{3}t(\hat{i} + \hat{j} + \hat{k})$$

when $t = 10 \text{ sec}$

$$r(10) = 20\sqrt{3}(10)(\hat{i} + \hat{j} + \hat{k})$$

$$r(10) = 200\sqrt{3}(\hat{i} + \hat{j} + \hat{k})$$

$$r(10) = (200\sqrt{3}, 200\sqrt{3}, 200\sqrt{3})$$

⑧ Distance from a point S to a line through P parallel to V .

$$d = \frac{|\overline{PS} \times V|}{|V|}$$

Example 5

$$S(1, 1, 5), P(1, 3, 0)$$

$$\vec{V} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\overline{PS} = -2\hat{j} + 5\hat{k}$$

$$\overline{PS} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = (-4+5)\hat{i} - \hat{j}(-5) + \hat{k}(2) \\ = \hat{i} + 5\hat{j} + 2\hat{k}$$

$$d = \frac{|\overline{PS} \times \vec{V}|}{|\vec{V}|} = \frac{\sqrt{1+25+4}}{\sqrt{1+1+4}} = \frac{\sqrt{30}}{\sqrt{6}}$$

$$\boxed{d = \sqrt{5}}$$

Equation for a plane

$$\underbrace{ax + by + cz}_{\text{Normal vector}} = d$$

$$\vec{n} = [a, b, c] = a\hat{i} + b\hat{j} + c\hat{k}$$

$$P(x, y, z), P_0(x_0, y_0, z_0)$$

$$\vec{P_0P} = (x-x_0)\hat{i} + (y-y_0)\hat{j} + (z-z_0)\hat{k}$$

$\vec{n} \cdot \vec{P_0P} = 0$ — (1) Vector form of equation of plane

$$(a\hat{i} + b\hat{j} + c\hat{k}) \cdot ((x-x_0)\hat{i} + (y-y_0)\hat{j} + (z-z_0)\hat{k}) = 0$$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0 \text{ — (2)}$$

Component equation.

$$ax - ax_0 + by - by_0 + cz - cz_0 = 0$$

$$ax + by + cz = ax_0 + by_0 + cz_0$$

$$ax + by + cz = d \text{ — (3)}$$

Ex 6 $P_0(-3, 0, 7) \perp n = 5\hat{i} + 2\hat{j} - \hat{k}$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$$5(x+3) + 2(y-0) - 1(z-7) = 0$$

$$5x + 15 + 2y - z + 7 = 0$$

$$5x + 2y - z = -22$$

$$\boxed{5x + 2y - z = -22}$$

$$A(0,0,1), B(0,2,0), C(3,0,0)$$

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & -1 \\ 3 & 0 & -1 \end{vmatrix}$$

$$= \hat{i}(-2) - \hat{j}(3) + \hat{k}(-6)$$

$$\vec{n} = -2\hat{i} - 3\hat{j} - 6\hat{k}$$

We substitute the components of this vector and the coordinates of $A(0,0,1)$ into the component form of the equation to obtain

$$-2(x-0) - 3(y-0) - 6(z-1) = 0$$

$$-2x - 3y - 6z + 6 = 0$$

$$-2x - 3y - 6z = -6$$

$$\boxed{2x + 3y + 6z = 6}$$

$$3(x-0) + 2(y-0) + 6(z-1) = 0$$

$$3x + 2y + 6z - 6 = 0$$

$$\boxed{3x + 2y + 6z = 6}$$

Example 17

$$A(0,0,1), B(2,0,0) \text{ and } C(0,3,0)$$

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix} = \hat{i}(3) - \hat{j}(-2) + \hat{k}(6)$$

We substitute the components of this vector

and the coordinate of $A(0,0,1)$ into the component form of equation to obtain

Lines of intersection

Example 8

$$3x - 6y - 2z = 15$$

$$2x + y - 2z = 5$$

$$n_1 \times n_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix}$$

$$= \hat{i}(12+2) - \hat{j}(-6+4) + \hat{k}(3+12)$$

$$= 14\hat{i} + 2\hat{j} + 15\hat{k}$$

The Distance from a point to a plane

Distance from a point S to a plane with Normal n at point P

$$d = \left| \overrightarrow{PS} \cdot \frac{n}{|n|} \right|$$

Example:- Find the distance from $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Solve:-

$$n = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

$$P = (0, 3, 0)$$

$$\overline{PS} = (1-0)\hat{i} + (1-3)\hat{j} + (3-0)\hat{k}$$

$$\overline{PS} = \hat{i} - 2\hat{j} + 3\hat{k}$$

$$|n| = \sqrt{(3)^2 + (2)^2 + (6)^2}$$

$$|n| = \sqrt{9+4+36} = \sqrt{49} = 7$$

Now,

$$d = \left| \overline{PS} \cdot \frac{n}{|n|} \right|$$

$$= \left| (\hat{i} - 2\hat{j} + 3\hat{k}) \cdot \left(\frac{3}{7}\hat{i} + \frac{2}{7}\hat{j} + \frac{6}{7}\hat{k} \right) \right|$$

$$= \left| \frac{3}{7} - \frac{4}{7} + \frac{18}{7} \right| = \boxed{\frac{17}{7}} \text{ Ans}$$

Example:- Find angles between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Sol:- The vectors

$$n_1 = 3\hat{i} - 6\hat{j} - 2\hat{k} = 15, \quad n_2 = 2\hat{i} + \hat{j} - 2\hat{k} = 5$$

$$\theta = \cos^{-1} \frac{n_1 \cdot n_2}{|n_1| |n_2|} = \cos^{-1} \left(\frac{4}{21} \right) \approx 1.38 \text{ radian}$$

Finding parametric equations for lines in Exercise 1-12

(1) The line through the point $P(x_0, y_0, z_0)$ parallel to the vector $\hat{i} + \hat{j} + \hat{k}$.

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$
$$x = 3 + t, \quad y = -4 + t, \quad z = -1 + t$$

(4) The line through $P(1, 2, 0)$ and $Q(1, 1, -1)$

$$\overrightarrow{PQ} = (1-1)\hat{i} + (1-2)\hat{j} + (-1-0)\hat{k}$$
$$\overrightarrow{PQ} = -\hat{j} - \hat{k}$$

$$x = x_0 + at = 1 + 0t = 1$$
$$y = y_0 + bt = 2 + (-1)t = 2 - t$$
$$z = z_0 + ct = 0 + (-1)t = -t$$

(9) The line through $(3, 2, -8)$ and is perpendicular to the plane $3x - y - 2z + 2 = 0$

$$P_0 = (3, 2, -8)$$

$$n = 3\hat{i} - \hat{j} - 2\hat{k} = -2$$

$$x = x_0 + at = 3 + 3t$$

$$y = y_0 + bt = 2 + (-1)t = 2 - t$$

$$z = z_0 + ct = -8 - 2t = -8 - 2t$$

(10) The line through $(2, 3, 0)$ perpendicular to the ~~plane~~ vectors $u = \hat{i} + 2\hat{j} + 3\hat{k}$ and $v = 3\hat{i} + 4\hat{j} + 5\hat{k}$

$$P_0 = (2, 3, 0)$$

For unit vector $\hat{n}$ we take $\overline{u \times v}$.

$$\begin{aligned}\overline{u \times v} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 4 & 5 \end{vmatrix} = \hat{i} \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 3 \\ 3 & 5 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \\ &= \hat{i}(10-12) - \hat{j}(5-9) + \hat{k}(4-6) \\ \overline{n} &= -2\hat{i} + 4\hat{j} - 2\hat{k}\end{aligned}$$

$$\overline{n} = (a, b, c) = (-2, 4, -2)$$

$$\begin{aligned}x &= x_0 + at = 2 - 2t \\ y &= y_0 + bt = 3 + 4t \\ z &= z_0 + ct = 0 - 2t = -2t\end{aligned}$$

Finding parametrization for the line segments

$$(13) \quad (0, 0, 0) \rightarrow (1, 1, 3/2)$$

$$P(0, 0, 0), Q(1, 1, 3/2)$$

$$\begin{aligned}\overline{PQ} &= (1-0)\hat{i} + (1-0)\hat{j} + (3/2-0)\hat{k} \\ &= \hat{i} + \hat{j} + 3/2\hat{k}\end{aligned}$$

$$\begin{aligned} x &= x_0 + at = 0 + 1t = t \\ y &= y_0 + bt = 0 + 1t = t \\ z &= z_0 + ct = 0 + 3t = 3t \end{aligned}$$

(19) $(2, 0, 2), (0, 2, 0)$

$$P_0 = (x_0, y_0, z_0) = (2, 0, 2)$$

$$Q = (0, 2, 0)$$

$$\overrightarrow{PQ} = (0-2)\hat{i} + (2-0)\hat{j} + (0-2)\hat{k}$$

$$\overrightarrow{PQ} = -2\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\begin{aligned} x &= x_0 + at = 2 - 2t \\ y &= y_0 + bt = 0 + 2t \\ z &= z_0 + ct = 2 - 2t \end{aligned}$$

* Finding equation of planes

(21) The plane through $P_0(0, 2, -1)$ normal to $n = 3\hat{i} - 2\hat{j} - \hat{k}$

$$P(x, y, z), P_0(0, 2, -1), n = (3\hat{i} - 2\hat{j} - \hat{k})$$

Equation of plane

$$a(x - x_0) + b(y - y_0) + c(z - z_0)$$

$$3(x - 0) - 2(y - 2) - 1(z + 1)$$

$$3(x - 0) - 2(y - 2) - 1(z + 1)$$

$$3x - 2y + 4 - z - 1 = 0$$

$$3x - 2y - z = -3$$

26) The plane through $A(1, -2, 1)$, perpendicular to the vector from the origin to A .

$$A(1, -2, 1), \quad P_0(0, 0, 0)$$

$$P(1, -2, 1), \quad n = \hat{i} - 2\hat{j} + \hat{k}$$

Equation of plane

$$a(x - x_0) + b(y - y_0) + c(z - z_0)$$

$$1(x - 1) + (-2)(y + 2) + 1(z - 1)$$

$$x - 1 - 2y - 4 + z - 1 = 0$$

$$x - 2y + z - 6 = 0$$

$$\boxed{x - 2y + z = 6}$$